**KATIE E. BRISTOL**
Department of Geological Sciences
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## EDUCATION

**Ph.D. in Geological Sciences** *In progress*
University of Florida, Gainesville, FL
Advisor: Dr. Courtney Sprain

**M.S. in Geophysics** 2020
Michigan Technological University, Houghton, MI
Advisor: Dr. Aleksey Smirnov

**B.S. in Applied Geophysics** w/ Geological Engineering minor 2018
Michigan Technological University, Houghton, MI
Field courses: University of Canterbury, NZ (Summer 2017)

## RESEARCH INTERESTS

Planetary science including rock magnetism, paleomagnetism, paleointensity, Earth history and geochronology, Solar System origin/evolution, and extraterrestrial mineral assemblages.

## RESEARCH AND WORK EXPERIENCE

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| Research Assistant, Neil Opdyke Paleomagnetics Lab, University of Florida | 2020 - present |
| Lab Manager, Earth Magnetism Lab, Michigan Tech. University | 2017 - 2020 |
| Research Assistant, Earth Magnetism Lab, Michigan Tech. University | 2015 - 2020 |
| NASA Michigan Space Grant Fellow, Michigan Tech. University | 2018 - 2019 |
| Summer Undergraduate Research Fellow, Michigan Tech. University | 2016 - 2017 |
| NSF Summer REU Intern, Michigan Tech. University | 2015 |

## PEER-REVIEWED PUBLICATIONS

1. Smirnov, A.V., E.V. Kulakov, M.S. Foucher, **K.E. Bristol**. "Intrinsic Paleointensity Bias and the Long-Term History of the Geodynamo," Science Advances, vol. 3, e1602306, 2017.

## SCIENTIFIC PRESENTATIONS AND ABSTRACTS

**** = presenting author***

1. Sprain*, C.J., E. Hurst, G.A. Paterson, R.K. Bono, **K.E. Bristol**, and A.J. Biggin, "Constraining the behavior of ε-$Fe_2O_3$ during paleointensity experimentation", Abstract ID#940005, American Geophysical Union 2021 Fall Meeting, New Orleans, 2021.
2. **Bristol*, K.E.,** C.J. Sprain, and J.G. Meert, "Absolute paleointensity for Precambrian-aged dikes from India", Abstract 227-11, Geological Society of America Abstracts with Programs, Vol. 53, No. 6, 2021.
3. Mijjum*, M., **K.E. Bristol**, M.M. Tremblay, and C.J. Sprain, "A model framework for cosmogenic nuclide production rates through deep time", Abstract 31-12, Geological Society of America Abstracts with Programs, Vol. 53, No. 6, 2021.
4. **Bristol*, K.E.**, E.J. Piispa, and A.V. Smirnov, "Thermomagnetic Behavior of Extraterrestrial Minerals: An Overview", Abstract GP004-05, American Geophysical Union 2020 Fall Meeting, Online Everywhere, 2020. (Invited oral)
5. Sanmartin*, K.J., E.J. Piispa, C.L. Mandon, M. Roverato, **K.E. Bristol**, A.V. Smirnov, and R. Trindade, "Rock Magnetism, Paleomagnetism, and Paleointensity of Imbabura Volcano (Ecuador) – Implications for the Spatiotemporal Growth Model", Abstract GP003-09, American Geophysical Union 2020 Fall Meeting, Online Everywhere, 2020.

6. **Bristol*, K.E.**, A.V. Smirnov, E.J. Piispa, A. Kosterov, E.V. Kulakov, and M. Ramirez, "Rock Magnetic Investigation of the Daule Ordinary Chondrite", Abstract GP014A-503767, American Geophysical Union 2019 Fall Meeting, San Francisco, 2019.
7. Piispa*, E.J., P. Larrea, K. Choez, **K.E. Bristol**, M. Roverato, A.V. Smirnov, and C. Mandon, "Paleo- and rock-magnetic record of the Imbabura volcanic units: Implications for the tectonomagmatic evolution of the volcano and for the Earth's magnetic field at equatorial latitudes," International Symposium on Andean Geodynamics, Quito, Ecuador, 2019.
8. Foucher, M.S., **K.E. Bristol***, E.J. Piispa, and A.V. Smirnov, "Absolute geomagnetic intensity from the ~1.1 Ga Baraga-Marquette Dike Swarm (Michigan, USA) obtained using the Shaw and pseudo-Thellier methods," Abstract GP21A-0633, American Geophysical Union 2018 Fall Meeting, Washington D.C., 2018.
9. Foucher*, M.S., E. Engel, **K.E. Bristol**, and A.V. Smirnov, "Evolution of large lava flows in rift setting: Paleomagnetic and rock magnetic insights into the Greenstone Flow," Abstract ME23C-082, IAVCEI 2017 Scientific American Assembly, Portland, OR, 2017.
10. **Bristol*, K.E.**, "Rock magnetic investigation of the carbonaceous chondrules from the Allende meteorite," Michigan Technological University Pavlis Honors College Undergraduate Research Symposium, Houghton, MI, 2017.

## TEACHING EXPERIENCE

Teaching Assistant, Michigan Tech. University
GE3040: Fundamentals of Geophysics — Spring 2020

Co-Instructor, Michigan Tech. University
GE4530/5340: Planetary Geology/Geophysics — Fall 2019

## OUTREACH AND MENTORING

**El = Elementary, HS = High School, UG = Undergraduate**

Exhibitor
9th Annual Elementary Science Fair, Duval County, FL — 2022
~600 El students attended

Instructor
Summer Youth Programs, Michigan Tech. University — 2015, 2016, & 2019
-Geological Engineering Exploration Program (~150 El + HS students)
-Women in Engineering Program (~450 El + HS students)
-Engineering Scholars Program (~420 El + HS students)

Mentor
Assistant Research Mentor, University of Florida — 2020 – present
2 UG students
Assistant Research Mentor, Michigan Tech. University — 2017 - 2020
1 HS, 2 UG
Assistant Mentor, NSF K-12 STEM Teacher Outreach, Michigan Tech. University — 2015
2 HS STEM teachers

## AWARDS AND HONORS

Graduate Student College of Liberal Arts & Sciences Travel Award, University of Florida — 2021
Awarded to present research at the Geological Society of America conference.

Gates Cambridge Scholarship **US Finalist**, Gates Cambridge Trust, University of Cambridge 2020
A full-cost scholarship to study at the University of Cambridge offered to intellectually outstanding postgraduate students. I was selected as a US finalist.

Dr. Nancy Scofield Pioneering Research Award 2019
Given to a graduate student whose work expands the boundaries of research in the Department of Geological and Mining Engineering and Sciences.

Geophysics Award, Michigan Tech. University 2019
Given for exceptional work in the field course GE3900: Field Geophysics

Departmental Scholar, Michigan Tech. University 2017
Given to an individual who best represents student scholarship in the Dept.

## PRESS RELEASES AND NEWS

"Old Rocks, Biased Data: Overcoming Challenges Studying the Geodynamo" 2017
Published by Michigan Tech News
Picked up by NSF, ScienceDaily, Phys, Newswise, and others

"Lab Tour: Earth Magnetism Laboratory" 2017
Published by Michigan Tech: Unscripted Research Blog

## FIELD EXPERIENCE

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| Custer National Forest, Montana – Sample collection | 1 week, 2021 |
| Hell Creek Region, Montana – Sample collection | 1 week, 2021 |
| Calumet, Michigan – Archeologic excavation | 2 days, 2018 |
| Keweenaw Peninsula, Michigan – Geophysical surveying | 5 weeks, 2018 |
| Midcontinent Rift, Michigan – Sample collection | Various dates, 2015 - 2018 |
| North & South Islands, New Zealand – Sample collection/mapping | 6 weeks, 2017 |